Restaurant Management Platform

Simple Application, Technology Stack & Module Documentation

This document provides a simple overview of the applications, technology stack, and major modules planned for the Restaurant Management Platform.

1. Applications

Application	Users	Main Purpose
Admin Web Application	Restaurant Admin / Manager	Restaurant configuration, menu, users, orders, reports and enabled modules.
POS Application	Cashier / Staff	Billing, payments, orders and table-related operations.
Captain / Waiter Application	Captain / Waiter	Take table orders and send KOT to the kitchen.
KOT Application	Service / Kitchen Staff	Create and manage Kitchen Order Tickets.
KDS Application	Kitchen Staff	Display and manage kitchen orders and preparation status.
Customer Web Website	Customers	Menu browsing, QR ordering, online ordering, payment, tracking and table booking

2. Application Model

One Platform → Many Restaurants → Different Subscriptions → Enabled Applications & Modules

Each restaurant can subscribe to the applications and modules it requires. The Admin Web Application, when included in a subscription, provides access to the restaurant's enabled features. Restaurant users only access the modules allowed by their roles and permissions.

3. Technology Stack

Layer	Technology	Purpose
Frontend	React + Vite	Build web applications and user interfaces.
Backend	Node.js + Express.js	Build REST APIs and handle business logic.
Database	PostgreSQL	Store restaurant, user, menu, order, payment, inventory and other business data

Basic Architecture

React + Vite → Node.js + Express.js APIs → PostgreSQL Database

4. Module List

Main Module	Key Sub-Modules
Restaurant Management	Restaurant profile, outlet, business hours, settings
User & Role Management	Users, roles, permissions, access control, audit
Subscription & Plan Management	Plans, subscriptions, enabled modules, limits
Menu Management	Categories, products, variants, modifiers, pricing, availability
Table Management	Floors, areas, tables, table status, QR
Reservation & Booking	Table availability, reservations, advance payment, cancellation
Order Management	Dine-in, takeaway, QR, website and delivery orders
POS & Billing	POS orders, bills, discounts, invoices, split bills
Payment Management	Cash, card, UPI/online payments, refunds, payment status
Captain / Waiter	Table selection, order taking, KOT submission, order status
KOT	Kitchen order tickets, items, status, reprint
KDS / Kitchen	Kitchen queue, preparation status, ready/completed orders
Inventory Management	Stock, stock transactions, issue, transfer, adjustment, consumption
Purchasing	Purchase request, purchase order, goods receiving, purchase return
Supplier Management	Supplier details, supplied items, purchase history, returns
Wastage Management	Wastage entry, reason, quantity, stock impact
Customer / CRM	Customer profile, order history, segments, feedback
Loyalty & Rewards	Points, rewards, redemption, loyalty history
Feedback & Complaints	Feedback, complaint, assignment, status, resolution
Staff & Shift Management	Employees, shifts, opening/closing cash, reconciliation
Expense Management	Expense categories, expenses, approvals, history
Tax Management	Tax rates, tax categories, product/service tax
Discount & Offers	Discounts, coupons, offers, limits, approvals
Online Ordering	Website menu, cart, online order, payment, tracking
QR Ordering	Table QR, menu, cart, order, payment / pay at counter
Delivery Management	Assignment, pickup, out-for-delivery, delivered
Delivery Integrations	Swiggy, Zomato and future delivery partners
Notifications	Customer, kitchen, staff and admin notifications
Reports & Analytics	Sales, payments, kitchen, inventory, customers, staff, expenses
Hardware	POS terminal, receipt printer, KOT printer, KDS, cash drawer

5. Main Business Flow

Customer / Staff → Order → KOT → Kitchen / KDS → Preparation → Serve / Pack → Bill → Payment → Completed Order → Reports

6. Inventory & Purchasing Flow

Purchase Request → Purchase Order → Supplier → Goods Receiving → Stock Increase → Kitchen Issue → Consumption → Wastage / Adjustment → Stock Balance

7. Customer Ordering Flow

Website / QR → Menu → Cart → Order → Payment (where required) → Order Management → KOT → Kitchen → Ready → Serve / Pickup

8. Simple Database Structure

PostgreSQL will use one database for the SaaS platform. Restaurant business data is separated using **restaurant_id**. For future multi-outlet support, **outlet_id** can be used where required.

Area	Example Tables
Restaurant	restaurants, outlets, settings
Users	users, roles, permissions, user_roles
Menu	categories, products, variants, modifiers
Orders	orders, order_items
Kitchen	kots, kot_items, kitchen_stations
Billing	bills, payments, refunds
Inventory	inventory_items, stock_transactions, wastage
Purchasing	purchase_orders, purchase_items, goods_receipts
Customers	customers, loyalty_transactions, feedback
Operations	employees, shifts, expenses, notifications

9. Development Scope Summary

• Frontend: React + Vite

• Backend: Node.js + Express.js

• Database: PostgreSQL

• Architecture: REST API based

• Platform Type: Subscription-based multi-restaurant SaaS

• Access: Application, module and role-based permissions

• Future Integration: Swiggy, Zomato and other delivery partners

10. Final Structure

Restaurant Platform
■■■ Admin Web
■■■ POS
■■■ Captain / Waiter
■■■ KOT
■■■ KDS
■■■ Customer Web Website